torch.kron#

https://pytorch.org/docs/stable/generated/torch.kron.html

Description:

Computes the Kronecker product, denoted by ⊗\otimes⊗, of input and other. If input is a (a0×a1×⋯×an)(a_0 \times a_1 \times \dots \times a_n)(a0×a1×⋯×an) tensor and other is a (b0×b1×⋯×bn)(b_0 \times b_1 \times \dots \times b_n)(b0×b1×⋯×bn) tensor, the result will be a (a0∗b0×a1∗b1×⋯×an∗bn)(a_0*b_0 \times a_1*b_1 \times \dots \times a_n*b_n)(a0∗b0×a1∗b1×⋯×an∗bn) tensor with the following entries: where kt=it∗bt+jtk_t = i_t * b_t + j_tkt=it∗bt+jt for 0≤t≤n0 \leq t \leq n0≤t≤n. If one tensor has fewer dimensions than the other it is unsqueezed until it has the same number of dimensions. Supports real-valued and complex-valued inputs.

Function Signature

```
torch.kron(input, other, *, out=None) → Tensor#
Parameters
Keyword Arguments
```

Parameters

Keyword

out (Tensor, optional) – The output tensor. Ignored if None. Default: None

Code Examples

```
>>> mat1 = torch.eye(2)
>>> mat2 = torch.ones(2, 2)
>>> torch.kron(mat1, mat2)
tensor([[1., 1., 0., 0.],
        [1., 1., 0., 0.],
        [0., 0., 1., 1.],
        [0., 0., 1., 1.]])

>>> mat1 = torch.eye(2)
>>> mat2 = torch.arange(1, 5).reshape(2, 2)
>>> torch.kron(mat1, mat2)
tensor([[1., 2., 0., 0.],
        [3., 4., 0., 0.],
        [0., 0., 1., 2.],
        [0., 0., 3., 4.]])
```

Notes & Warnings

NOTE

Note This function generalizes the typical definition of the Kronecker product for two matrices to two tensors, as described above. When input is a (m×n)(m \times n)(m×n) matrix and other is a (p×q)(p \times q)(p×q) matrix, the result will be a (p∗m×q∗n)(p*m \times q*n)(p∗m×q∗n) block matrix: A⊗B=[a11B⋯a1nB ⋮ ⋱ ⋮ am1B⋯amnB]\mathbf{A} \otimes \mathbf{B}=\begin{bmatrix} a_{11} \mathbf{B} & \cdots & a_{1 n} \mathbf{B} \\ \vdots & \ddots & \vdots \\ a_{m 1} \mathbf{B} & \cdots & a_{m n} \mathbf{B} \end{bmatrix} A⊗B=a11B ⋮ am1B⋯⋱⋯a1nB ⋮ amnB where input is A\mathbf{A}A and other is B\mathbf{B}B.

Detailed Documentation

Documentation

Computes the Kronecker product, denoted by ⊗\otimes⊗, of input and other.

If input is a (a0×a1×⋯×an)(a_0 \times a_1 \times \dots \times a_n)(a0×a1×⋯×an) tensor and other is a (b0×b1×⋯×bn)(b_0 \times b_1 \times \dots \times b_n)(b0×b1×⋯×bn) tensor, the result will be a (a0∗b0×a1∗b1×⋯×an∗bn)(a_0*b_0 \times a_1*b_1 \times \dots \times a_n*b_n)(a0∗b0×a1∗b1×⋯×an∗bn) tensor with the following entries:

where kt=it∗bt+jtk_t = i_t * b_t + j_tkt=it∗bt+jt for 0≤t≤n0 \leq t \leq n0≤t≤n. If one tensor has fewer dimensions than the other it is unsqueezed until it has the same number of dimensions.

Supports real-valued and complex-valued inputs.

This function generalizes the typical definition of the Kronecker product for two matrices to two tensors, as described above. When input is a (m×n)(m \times n)(m×n) matrix and other is a (p×q)(p \times q)(p×q) matrix, the result will be a (p∗m×q∗n)(p*m \times q*n)(p∗m×q∗n) block matrix:

where input is A\mathbf{A}A and other is B\mathbf{B}B.

out (Tensor, optional) – The output tensor. Ignored if None. Default: None